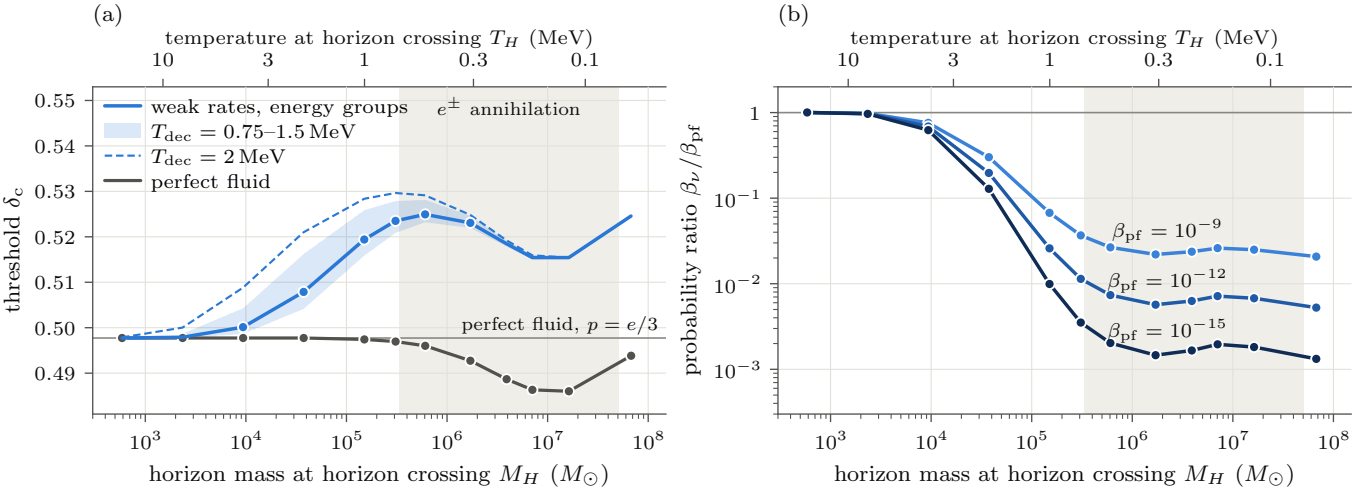

(a)
temperature at horizon crossing T_H (MeV)
10
3
1
0.3
0.1
weak rates, energy groups
$e^{\pm}$ annihilation
$T_{\rm dec} = 0.75$–$1.5\,{\rm MeV}$
$T_{\rm dec} = 2\,{\rm MeV}$
perfect fluid
perfect fluid, $p = e/3$
threshold $\delta_{\rm c}$
0.55
0.54
0.53
0.52
0.51
0.50
0.49
10^3
10^4
10^5
10^6
10^7
10^8
horizon mass at horizon crossing M_H ($M_\odot$)
(b)
temperature at horizon crossing T_H (MeV)
10
3
1
0.3
0.1
probability ratio $\beta_\nu/\beta_{\rm pf}$
1
10^{-1}
10^{-2}
10^{-3}
$\beta_{\rm pf} = 10^{-9}$
$\beta_{\rm pf} = 10^{-12}$
$\beta_{\rm pf} = 10^{-15}$
10^3
10^4
10^5
10^6
10^7
10^8
horizon mass at horizon crossing M_H ($M_\odot$)